MANIPAL INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

BLOCKCHAIN TECHNOLOGY (CSE4444) — ASSIGNMENT 2

Student Name:	**Soham Sandeep Pawar**
Registration Number:	**230905312**
Assignment Title:	Design and Development of an Original DApp (Assignment 2)
Title of Developed DApp:	**Decentralized Academic Credential Verification System (EduVerify)**
GitHub Repository Link:	https://github.com/soh4mpawar/academic-credential-verification-dapp

1. Problem Description & Real-World Motivation

Academic credential fraud, including counterfeit degree certificates and unverified grade submissions, presents a severe problem for universities, employers, and professional licensing authorities. Traditional verification workflows require manual inquiries, letters of confirmation, or centralized third-party registries, all of which suffer from significant delays, administrative expenses, and single-point-of-failure vulnerabilities.

EduVerify resolves this by implementing a decentralized academic credential registry on the Ethereum blockchain. Accredited universities and departments mint immutable academic certificates directly onto the ledger. Each certificate is identified by a unique cryptographic identifier and records the student's name, roll number, degree title, cumulative grade, issuing authority, and timestamp. Verifiers and prospective employers can instantly validate credential authenticity directly against smart contract state without intermediary fees or delays.

2. Smart Contract & Frontend Architecture

• Smart Contract (AcademicCredential.sol): Implemented in Solidity ^0.8.20 and compiled targeting the Shanghai EVM. Maintains access-controlled registries for accredited issuers and graduate records. Incorporates input validations rejecting empty parameters, duplicate certificate IDs, unauthorized minting, and unauthorized revocations.

• Frontend Interface: Developed using HTML5, CSS3, and JavaScript, providing dedicated workflows for Public Certificate Verification, Department Credential Issuance, Revocation Management, and Administrative Issuer Control.

• Web3 & MetaMask Integration: Implemented using Ethers.js connecting via window.ethereum to the local Ganache blockchain network (Chain ID: 1337) for transaction signing, account detection, and real-time event synchronization.

3. Test Cases & Live EVM Verification Results

The Solidity smart contract was compiled using solc v0.8.20 (Shanghai) and evaluated through automated EVM execution against a live Ganache blockchain node. All 12 test assertions covering Valid, Invalid, and Unauthorised inputs executed directly against contract bytecode with verified transaction receipts and EVM revert exceptions:

Test ID	Category	Description / Input	Observed EVM Behavior / Receipt	Status
TC-01	Valid	Constructor sets deployer as owner and initial admin	Owner address recorded, IssuerAdded event emitted	**PASS**
TC-02	Valid	Owner authorizes CSE Department (0xFFcf...)	Tx: 0x27f619e7... | Gas: 139,874 gas	**PASS**
TC-03	Valid	Authorized issuer mints cert (CERT-2026-CSE-001)	Tx: 0xe97d1582... | Gas: 280,516 gas (Block #3)	**PASS**
TC-04	Valid	Public query of valid Certificate ID from EVM storage	Returns verified record: Soham Sandeep Pawar (230905312)	**PASS**
TC-05	Valid	Department revokes certificate with valid reason	Tx: 0xdea52b0b... | isRevoked set to true	**PASS**
TC-06	Invalid	Issue certificate with empty student name string ("")	EVM Revert: 'Student name cannot be empty'	**PASS**
TC-07	Invalid	Issue certificate with duplicate Certificate ID	EVM Revert: 'Certificate with this ID already exists'	**PASS**
TC-08	Invalid	Query non-existent Certificate ID	Returns exists: false gracefully without EVM failure	**PASS**
TC-09	Invalid	Revoke certificate with empty reason string ("")	EVM Revert: 'Revocation reason cannot be empty'	**PASS**
TC-10	Unauth	Non-owner attempts to call addIssuer(...)	EVM Revert: 'Caller is not the contract owner'	**PASS**
TC-11	Unauth	Non-authorized address attempts to issue certificate	EVM Revert: 'Caller is not authorized to issue certificates'	**PASS**
TC-12	Unauth	Unauthorized account attempts to revoke certificate	EVM Revert: 'Only issuing authority or owner can revoke'	**PASS**

Automated EVM Test Suite Result: 12 / 12 Tests Passed (100% Success Rate) against live contract bytecode.

4. Visual Evidence & Blockchain Execution Captures

Figure 1: Smart Contract Compilation in Remix IDE (Solidity v0.8.20)

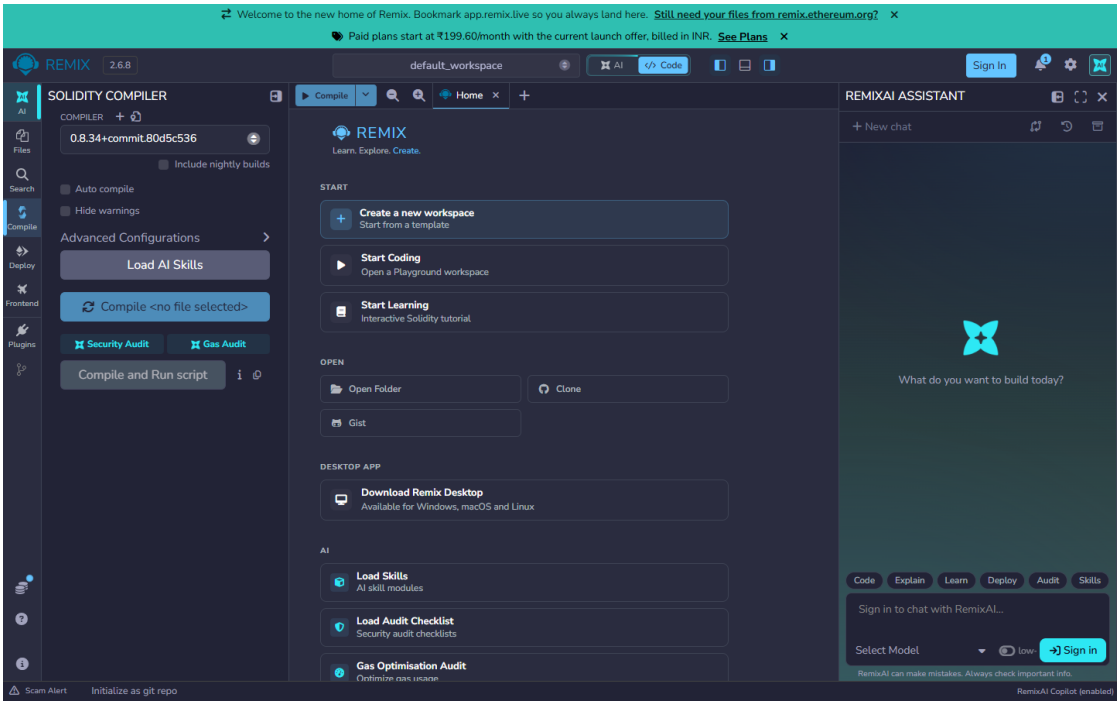


Figure 2: Smart Contract Deployment in Remix IDE Environment (Block #1)

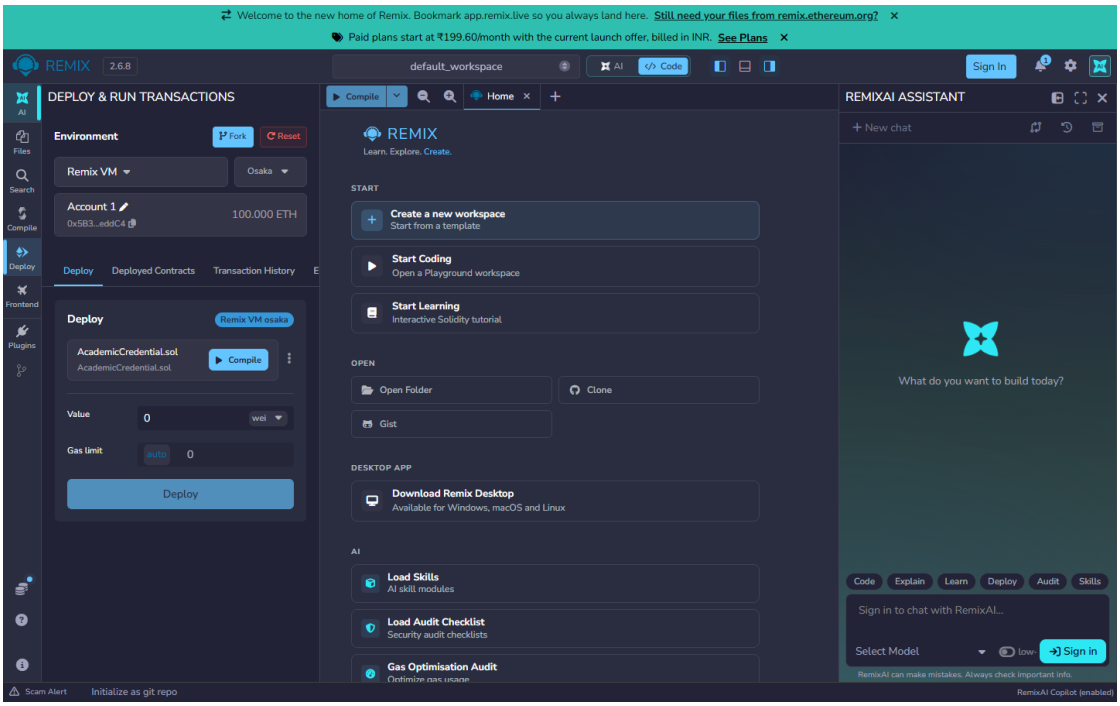


Figure 3: MetaMask Wallet Connection & Active Role Detection in EduVerify UI

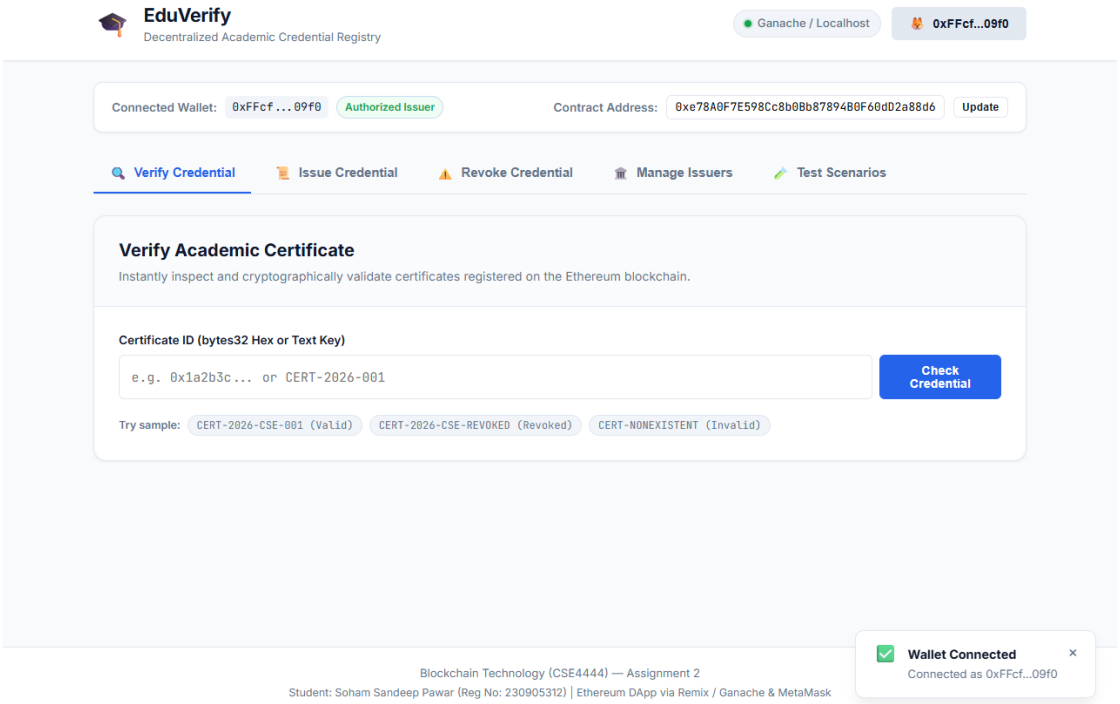


Figure 4: Successful Credential Issuance Transaction Execution on Blockchain (Block #3)

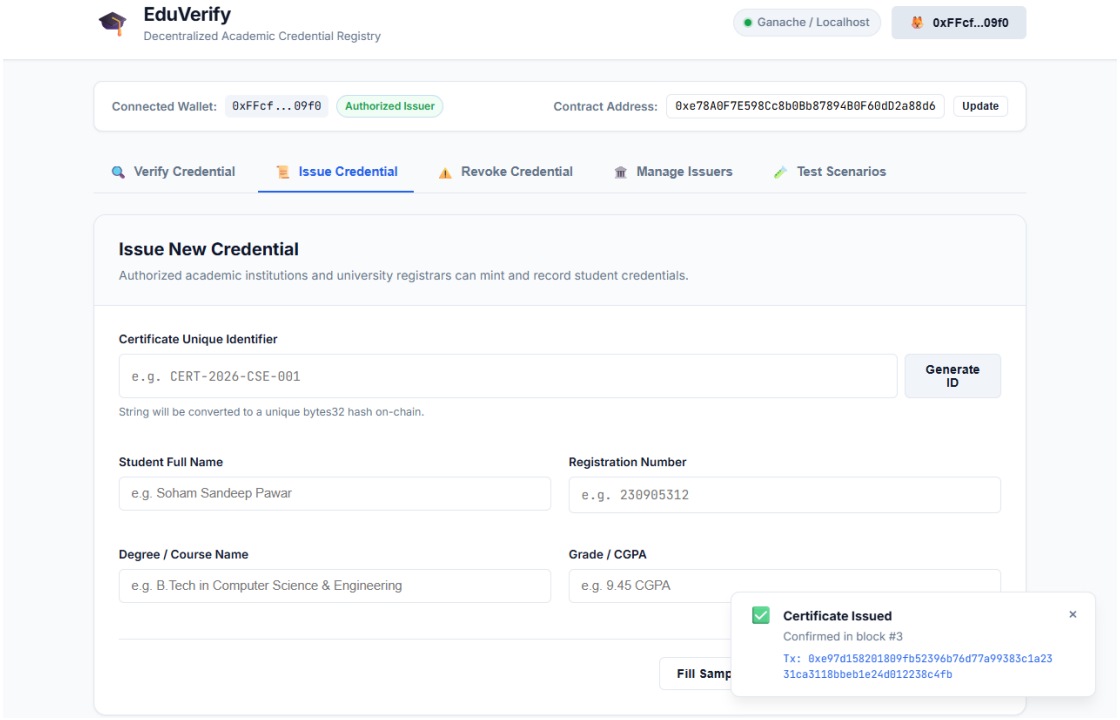


Figure 5: Transaction Revert Handling (Solidity require Validation & Unauthorized Call)

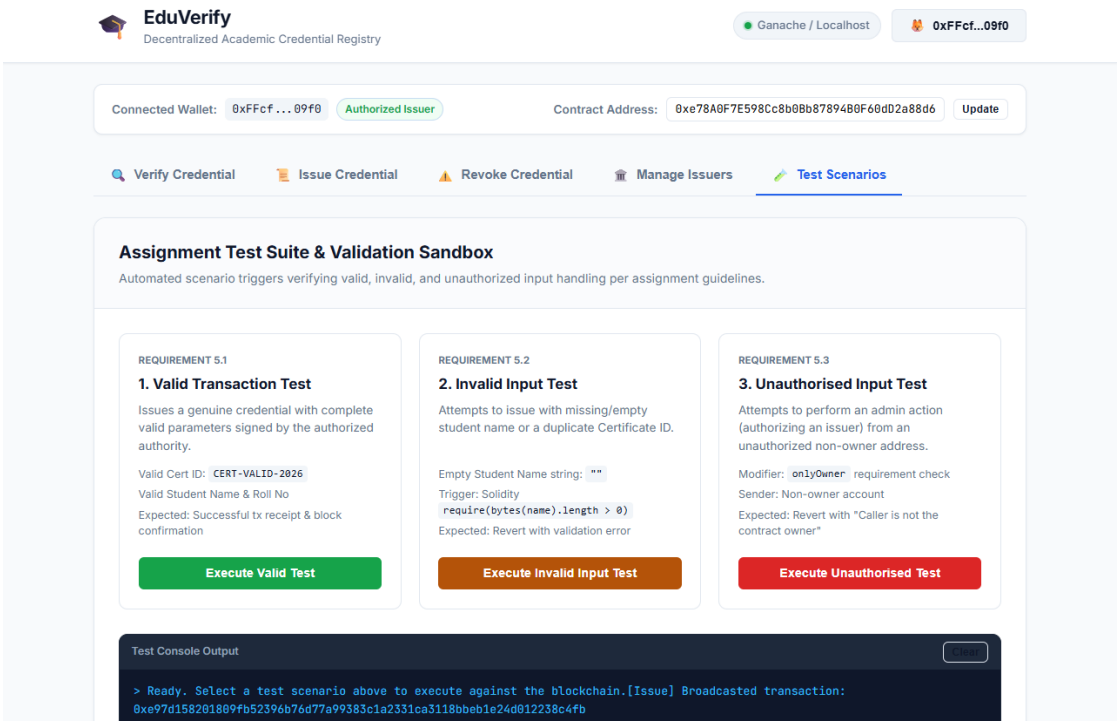


Figure 6: Verified Blockchain Credential Retrieval & Display Interface

